

Education

Northeastern University, Khoury College of Computer Science

Boston, MA

Bachelor of Science in Data Science and Mathematics

September 2024 – Expected May 2028

Data Science Concentration: Computer Science

GPA: 3.67/4.0 | Dean's List

Relevant Courses/Clubs: Machine Learning, Databases, Large-Scale Storage and Retrieval, Data Visualization

Skills/Interests

Languages: Java, Python (Numpy, Pandas), JavaScript, TypeScript, R, MySQL, VBA, DrRacket,

Technologies / Frameworks: Git, Docker, Spring Boot, React, Swing, [Node.js](#), Power BI, MATLAB

Development Practices: Test-Driven Development (JUnit, Vitest, Supertest), MVC Architecture, REST APIs

Specializations/Data Science: Statistical Analysis, Data Visualization, Data Cleaning & Processing

Interests: Full-Stack Development, Open-Source Contribution, User Experience Design, Billiards, Badminton

Work Experience

City of Quincy, Inspectional Services | *Excel VBA, OpenGov, Data Analysis, Power BI*

Quincy, MA

Summer Intern

July 2022-August 2023

- Optimized municipal permit processing workflow by extracting and analyzing 10,000+ permit records from the OpenGov system, implementing data cleaning protocols and compliance pattern analysis that reduced processing time by 40% and improved departmental efficiency
- Enhanced citizen services and regulatory compliance by processing 100+ permit applications through comprehensive blueprint review, building code verification, and project progress tracking while serving as Chinese interpreter for 30+ non-English speaking applicants, improving accessibility and approval timelines
- Automated administrative operations and enhanced revenue tracking visibility by developing 10+ VBA macros for comprehensive report generation, enabling management to monitor fee revenue streams, track inspector productivity metrics, and assess departmental progress while eliminating processing bottlenecks and reducing manual errors across multiple permit categories

Project Experience

Diabetes Data Analysis & Visualization Website | *JavaScript, D3.js, HTML, CSS*

Oct 2025-Dec 2025

- Investigated patterns in diabetic patient readmissions to help healthcare administrators reduce preventable costs
- Built interactive data visualization platform using D3.js to analyze survey data of 10,000+ diabetic patient readmissions; implemented dynamic scatterplots with filtering capabilities to explore correlations between patient demographics, medication regimens, hospital stay duration, and readmission likelihood
- Enabled healthcare stakeholders to identify high-risk patient populations and intervention opportunities through visual analysis of readmission patterns across age groups, medication types, and prior admission history

Settings Management System | *Node.js, Express, React, SQLite, Docker*

Jan 2026-Jan 2026

- Identified that non-technical users frequently introduced JSON syntax errors when editing raw configuration data, causing API failures and frustration.
- Designed and built a visual block-based JSON editor inspired by Scratch/Code.org, featuring type-specific input controls, real-time inline validation with contextual error messages, and dynamic field add/remove functionality, while rendering JSON syntax elements as non-editable visual components.
- Achieved comprehensive API test coverage across all five REST endpoints including edge cases with fully isolated, parallelizable tests using Vitest and Supertest.

Carnegie Mellon University, Computer Science Department | *Machine Learning*

Aug 2023-Oct 2023

- Supported Music AI research by organizing and preprocessing large-scale audio datasets for machine learning model training, contributing to synthetic voice and music generation technology development through systematic data labeling, voice pattern recognition, and quality assurance protocols
- Advanced AI voice system development through comprehensive audio analysis and feature extraction including lyrics extraction, audio segmentation, phoneme timing annotation, and cross-genre feature study, enabling improved voice cloning and music generation algorithms with enhanced accuracy
- Enhanced training dataset quality by contributing 500+ processed soundtrack samples with systematic data preparation and audio analysis, directly supporting AI music generation and voice synthesis research